

# Тригонометрические тождества, суммы и произведения

1 сентября 2026 года

## Теоретический минимум.

**Опр.** Every trigonometric computation in this sheet is one of three moves: pass to the complex exponential  $e^{ix} = \cos x + i \sin x$  and sum a geometric progression; telescope, having written the general term as a difference; or recognise the values  $\cos \frac{2\pi k}{n}$  as the roots of an explicit polynomial.

**Теорема 1 (ядро Дирихле)** For  $x \notin 2\pi\mathbb{Z}$ ,

$$\sum_{k=1}^n \cos kx = \frac{\sin \frac{nx}{2} \cos \frac{(n+1)x}{2}}{\sin \frac{x}{2}}, \quad \sum_{k=1}^n \sin kx = \frac{\sin \frac{nx}{2} \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}.$$

In particular both partial sums are bounded by  $1/|\sin \frac{x}{2}|$  uniformly in  $n$ .

**Теорема 2 (формула удвоения Виета)**  $\prod_{k=1}^n \cos \frac{x}{2^k} = \frac{\sin x}{2^n \sin \frac{x}{2^n}}$ , whence

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2}} \cdots$$

**Факт 1**  $\sum_{k=1}^n \tan^2 \frac{k\pi}{2n+1} = n(2n+1)$  and  $\sum_{k=1}^{n-1} \cot^2 \frac{k\pi}{n} = \frac{(n-1)(n-2)}{3}$ .

both are Vieta's formulas for a polynomial whose roots are those cotangents.

- 
1. Prove both identities of Theorem 1 in two ways: by telescoping  $2 \sin \frac{x}{2} \cos kx$  and by summing the geometric progression  $\sum e^{ikx}$ . Deduce that  $\sum_{k=1}^{\infty} \frac{\sin kx}{k}$  converges for every real  $x$  and equals  $\frac{\pi-x}{2}$  on  $(0, 2\pi)$ .
  2. Prove Theorem 2 and deduce Vieta's product for  $2/\pi$ . Then compute  $\prod_{k=1}^n \cos \frac{k\pi}{2n+1}$  and  $\prod_{k=1}^{n-1} 2 \sin \frac{k\pi}{n}$ .
  3. Prove that  $\sum_{k=1}^n \tan^2 \frac{k\pi}{2n+1} = n(2n+1)$ , and deduce the estimate  $\frac{\pi^2}{6} = \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \cot^2 \frac{k\pi}{2n+1}$  together with the classical proof of  $\sum 1/k^2 = \pi^2/6$ . (Подсказка:  $\cot^2 t < 1/t^2 < 1 + \cot^2 t$  на  $(0, \pi/2)$ .)

4. (Proposed problems, №267) Prove that  $\tan \frac{n\pi}{2n+1} - 2 \tan \frac{(n-1)\pi}{2n+1} > \frac{\pi}{2}$  for every positive integer  $n$ .
5. (Mexmat, 2001) Let  $P_n$  and  $Q_n$  be regular  $n$ -gons,  $P_n$  inscribed in a circle of diameter 1 and  $Q_n$  circumscribed about it, with perimeters  $p_n$  and  $q_n$ . In which third of the interval  $(p_n, q_n)$  does the number  $\pi$  lie?